CSE 216 Formal Methods
Fall 2026 Daniel Fremont

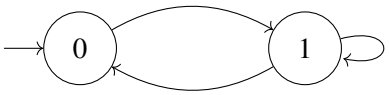
HW 5

Due November 3 at 3:00 pm

(1 question, 45 points total)

1. (45 pts.) Transition systems

Consider the following closed transition system M:



(a) What are all runs of M of length 3? (You can write them as triples of states.)

Answer:

(b) Let φ be the trace property "state 0 is visited infinitely many times". Does $M \models \varphi$? Why or why not?

Answer:

(c) Let's run the BDD-based algorithm for computing the reachable states of M. First, write a BDD for the function $S_0(s)$ representing the initial states, which will also be the BDD $R_0(s)$ of states reachable in at most 0 steps.

Answer:

(d) Write a BDD for the function $\delta(s,s')$ representing the transition relation.

Answer:

(e) Write a BDD for the function $R_0(s) \wedge \delta(s,s')$.

Answer:

(f) Write a BDD for the function $\exists s.(S_0(s) \wedge \delta(s,s'))$ (which should only depend on s').

Answer:

(g) Write a BDD for the function $R_1(s') = R_0(s) \vee \exists s.(S_0(s) \wedge \delta(s,s'))$.

Answer:

(h) What will happen in the next iteration of the algorithm?

Answer: